# Hands-On Activity: COUNT and COUNT DISTINCT

**1.**

Question 1

## Reflection

What is the key difference between `COUNT` and `COUNT DISTINCT` in a database query?

Do you understand?.

`COUNT` returns the number of records returned by a query. `COUNT DISTINCT` only returns the number of the distinct records that are returned by the query.

`COUNT` returns the number of values returned by a query. `COUNT DISTINCT` returns all values in a specified range.

`COUNT` returns the number of columns returned by a query. `COUNT DISTINCT` only returns the distinct values within those columns.

`COUNT` returns the sum of the values in a specified range. `COUNT DISTINCT` returns the number of rows in a specified range.

**Correct**

Answer: `COUNT` returns the number of records returned by a query. `COUNT DISTINCT` only returns the number of the distinct records that are returned by the query.

**2.**

Question 2

In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions.

- What are some of the benefits of using `COUNT` and `COUNT DISTINCT` when working with larger datasets?
- In which ways are the SQL and spreadsheet functions that you've been learning similar?

Do you understand?.

My Answer: COUNT returns the number of records by a query, and COUNT DISTINCT returns the same as the COUNT command but removes the duplicate value.

In the spreadsheet, there is also a count function, which counts the entire record value of a particular column, and the SQL count command does exactly this. Similarly, COUNT DISTINCT returns the same as the COUNT command but removes the duplicate value, and in the spreadsheet, we use the Remove Duplicates feature to remove the duplicate values. That's it.

**Feedback**

Great work completing this activity! In this activity, you executed queries using `COUNT` and `COUNT DISTINCT`, as well as `SELECT`, `FROM`, and `GROUP BY` statements to create aliases and `JOIN` tables, return numerical values within a specific range, and group by specific columns within a table. An effective response might include that `COUNT` and `COUNT DISTINCT` can help you answer questions about how many in a dataset or identify the number of non-null or null values in a specified range.